# Sherif Morsi

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## Summary

Computer Science student building **LLM agent systems**: multi-agent teams that talk in schema-checked JSON, signed mandates that bound what agents may do, **QLoRA** fine-tunes, and a **transformer** trained from scratch, each scored on a frozen eval.

## Education

**Concordia University** Montreal, QC, Canada
**Bachelor of Computer Science** *Expected May 2028*

- **Availability:** Winter 2027 (Jan to Apr 2027) for 4, 8, 12, or 16 months, anywhere in Canada
- **Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Artificial Intelligence, Machine Learning, Linear Algebra, Multivariable Calculus

## Experience

**AI & Machine Learning Intern** Aug 2026 - Sep 2026
**Nokia** *Cairo, Egypt*

- Trained and evaluated Python and **PyTorch** models on wireless network telemetry to forecast 5G **energy demand**
- Developed **data and ML pipelines** for 5G energy telemetry, to support automated network management and optimization
- Trained in **agentic AI** for intent-based 5G network slicing (Amazon Bedrock); presented findings to cross-functional data science and engineering teams and led the project proposal and requirements review

**Credit Analyst Job Shadow** Aug 2025 - Sep 2025
**Qatar National Bank (QNB)** *Cairo, Egypt*

- Built and documented **Python** utilities and **SQL** queries that preprocess and validate SME financial records for the bank's credit-review process, improving data quality and reliability

## Projects

**Aurvane (Agent Security, Delegation)** | *Python, Ed25519, MCP, JavaScript* | **GitHub** | **Live Demo** 2026

- Built a trust layer between LLM agents and their tools: **Ed25519-signed mandates**, four checks on every action (identity, scope, bounds, provenance), a human Allow/Deny step when untrusted text picks a destination, and a hash-chained audit log
- **Agent-to-agent delegation:** sub-agents only receive strictly narrower mandates that verify back to the human's key. Ships as a Python SDK decorator and an **MCP server** for Claude Code or any MCP client; 46 tests in CI
- Against a simulated agent that obeys every **prompt injection** (60 paired tasks), attacks that landed fell from **30/30 to 7/30** and money lost from 12,135 to **135 CAD**, with 0 false alarms; a 12-attack red team: 8 contained, 2 capped, 2 through

**Orbit (Multi-Agent Systems, Fine-Tuning)** | *Python, PyTorch, QLoRA, Claude API* | **GitHub** | **Live Replay** 2026

- Built a planner, coder, test runner, critic, and reviewer that talk in **schema-validated JSON** with per-role tool permissions; a code-level reviewer rejects patches that fail hidden tests or edit them, and every turn is a replayable JSONL trace
- **QLoRA**-tuned **Qwen2.5-1.5B** on agent dialogues: schema-valid tool-call rate **0.333 to 0.764** (95% CI 0.264 to 0.607). Task success 4/12 vs 2/12 (CI includes zero)
- On 32 frozen bug-fix tasks with Claude Haiku, the team passed **29/32** and a single agent 27/32, for **$0.018**; the paired 95% CI (0.000 to 0.156) includes zero. One model interface over five backends

**Time-Series Foundation Model (Transformers)** | *Python, PyTorch, pytest, GitHub Actions* | **GitHub** 2026

- Pretrained a **10.1M-parameter** decoder-only transformer from scratch on **443,258** series across 77 datasets. On the frozen test (97 configurations), CRPS was **0.688**× AutoARIMA, 95% CI [0.624, 0.757]
- A leakage audit caught a public benchmark corpus containing its own test data and removed **801 of 447,229** series before training; 170 automated tests in CI

**World Shock Lab (Econometrics, ML)** | *Python, PyTorch, DuckDB, TypeScript* | **GitHub** | **Live Site** 2026

- Estimated how Fed, Bank of Canada, and OPEC surprises move **53** assets (a 10 bp Fed surprise moves the TSX $-$**1.10%**, 90% band $-1.59$ to $-0.61$), replicated **11 of 13** published findings, and publishes forecasts before each decision

**ATLAS (Production Networks)** | *Python, PyTorch, FastAPI* | **GitHub** 2026

- Traced cost shocks through **405** country-sector nodes in **11.0 ms**; trained 12 PyTorch graph and no-graph checkpoints, and ridge still won on the held-out year (test RMSE 14.00)

## Technical Skills

**AI Agents / LLMs**: **LLM Agents**, **Multi-Agent Systems**, **Tool Calling**, Structured Outputs, **MCP**, Prompt Engineering, Prompt-Injection Defense, **QLoRA**, Distillation, LLM Evaluation, Claude API, Hugging Face
**Machine Learning**: **PyTorch**, TensorFlow, **scikit-learn**, **Transformers**, Time-Series Forecasting, Feature Engineering, Cross-Validation, Bootstrap CIs, Mixed Precision
**Languages**: **Python**, **SQL**, TypeScript, JavaScript, C++, Java, C, R, Bash
**Data / Backend**: **Pandas**, **NumPy**, SciPy, **FastAPI**, DuckDB, SQLite, Parquet, statsmodels, Econometrics
**Developer Tools**: **Git**, **GitHub Actions (CI/CD)**, **pytest**, Playwright, Docker, Jupyter, Colab, Claude Code, Cursor

## Certifications & Affiliations

**Anthropic:** Building with the Claude API, Claude Code 101, Claude 101 | **IBM:** Machine Learning with Python | **QUARCC** (Quantitative Research and Competitions Club), Member